Assignment 8

Exercise 3 : Is the ternary code {00,012,0110,0112,100,201,212,22 }uniquely decodable??

To determine if the ternary code {00,012,0110,0112,100,201,212,22} is uniquely decodable, we need to check if there is any prefix ambiguity in the code. A code is prefix-free, if no codeword is a prefix of another codeword.

The codewords are:

- 0000
- 012012
- 01100110
- 01120112
- 100100
- 201201
- 212212
- 2222

To check prefix Ambiguity :We need to verify whether any of these codewords is a prefix of another.

Let's compare each codeword with every other codeword:

1. "00": we need to check if it is a prefix of any other codeword.

   It is a prefix of 01100110, 01120112, and 100100.

Since 00 is a prefix of some other codewords, the code is not prefix-free.

So, the given ternary code is not uniquely decodable because 00 is a prefix of other codewords. Therefore, code has prefix ambiguity, and it's not a prefix-free code.

Exercise 4. Make Huffman codes for X2, X3 and X4 where AX ={0, 1} and PX = {0.9, 0.1}. Compute their expected lengths and compare them with the entropies H(X2), H(X3) and H(X4). Repeat this exercise for X2 and X4 where PX = {0.6, 0.4}.

Step 1. Huffman Coding for X2,X3,X4

Defining each case with the given probabilities and the alphabet AX ={0,1}

For X2, PX ={0.9,0.1}

For X3, PX ={0.9,0.1}

For X4, PX ={0.9,0.1}

Similarly, repeat for PX ={0.6,0.4}

Step 2. Huffman Code Construction :

Huffman coding assigns binary codes to symbols such that the expected length of the encoded message is minimized. The algorithm works by building a binary tree based on the probabilities of symbols, where less probable symbols are assigned longer codes, and more probable symbols are assigned shorter codes.

Step 3. Expected Length Calculation: The expected length L of the code is the weighted average length of the codewords:

L =P0*Length of code for 0 + P1*Length of code for 1

L =(0.9*1)+(0.1*1) = 1

The expected length L is **1**.

Step 4 :The entropy H(X) of a random variable with probabilities P0 and P1 is given by:

For PX = {0.9, 0.1} P0=0.9 and P1 =0.1

H(X)=−(0.9log 2 0.9 + 0.1log 0.1)

log2 0.9 ≈ −0.137

log2 0.1 ≈ -3.322

Step 5 :Substitute these values we get

$H(X)=-(0.9*(-0.137) + 0.1 *(-3.322))$

$H(X) = 0.468$

- The expected length of the Huffman code for PX={0.9,0.1}PX={0.9,0.1} is **1**.
- The entropy of the distribution PX={0.9,0.1}PX={0.9,0.1} is approximately **0.468**.

Conclusion : The code does not achieve the theoretical limit (the entropy), but this is common in simple coding schemes where the alphabet is small, and the code must assign a fixed-length codeword to each symbol.